ML pipeline (1/2)

MADRE: Méthodologie pour l'Analyse de Données Réseaux
(Methodology for Network Data Analysis)

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Outline

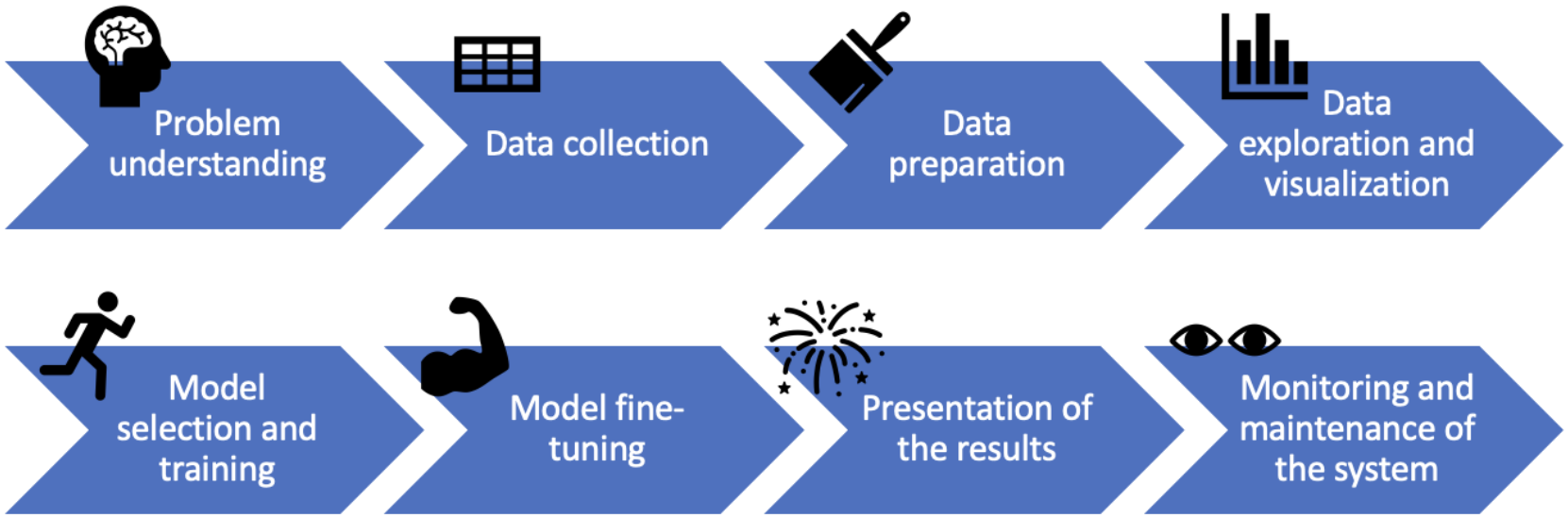
1. Introduction
2. Problem understanding
3. Data collection
4. Data preparation
5. Data exploration and visualization
6. Let's play

Introduction

Different actors are involved in a machine learning process:

- **Clients** have a specific need and require a project that uses ML.
- **Programmers** are programming sophisticated mathematical algorithms.
- **Machine learning engineers** tend to talk about model training and data wrangling.
- **Data engineers** works on large amounts of data with the objective to develop tools to make use of this data.
- ...

High level overview



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Problem understanding

The process of deeply understanding the real-world problem you are trying to solve with machine learning, before jumping into data collection, modeling, or algorithms.

It is a fundamental step: if you do not get this right, the whole ML project can head in the wrong direction.

Furthermore, it aligns the ML solution with the goals; prevents wasting time on irrelevant data or wrong modeling approaches; helps with effective communication across teams; sets the scope for the rest of the ML pipeline.

Key aspects

- **Identify the Real-World Objective**
  What is the actual problem you are solving?
  Why is this problem important to solve?
- **Define the ML Problem Type**
  Is it a classification, regression, clustering, etc.?
  What will the model do, in practical terms?
- **Clarify the Expected Output**
  What kind of prediction or decision do you expect?
  Is the output categorical, numerical, a ranking, a recommendation, etc.?
- **Understand the Evaluation Metrics**
  How will you measure the model's success?
  Is there a cost associated with errors?
- **Determine Constraints and Limitations**
  Are there privacy concerns?
  Do you have limited data?
  Are there latency, interpretability, or resource constraints?

Example

A company runs a large-scale network and wants to detect network intrusions (cyberattacks) in real-time.

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Why is this problem important to solve?
Manual monitoring is too slow and prone to error. Fast detection can prevent data breaches, service outages, and financial loss.

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- Binary classification: 0 if normal traffic; 1 if Malicous/Intrusion traffic;
- Multiclass classification: Classify specific types of attacks (e.g., DoS (Denial of Service), Port scanning, etc.).

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What will the model do, in practical terms?

- Monitor incoming network data (e.g., from a router, firewall, or packet capture tool);
- Extract features (e.g., bytes sent, duration, protocol, source/destination IP);
- Train and test a ML model.

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The output is categorical: Binary or multiclass categorical output.

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Is there a cost associated with errors?

Yes.

- False Negative: we do not detect an attack

  HIGH COST (downtime, damage to reputation, ...)

- False positive: We flag normal traffic as attack

  MEDIUM COST (false alarms, waste of resources, service disruption, ...)

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Latency? Yes, real-time detection requires fast, lightweight models.
Interpretability? Yes, we shouls trust and understand the classification
Resource? Possibily yes, we may need to deploy on limited hardware or edge devices.

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Data collection

The Data Collection is the phase in which the raw data that your model will learn from are gathered. It is a crucial step because the quality, quantity, and relevance of the data directly impact the model's performance. This phase includes gathering, sourcing and storing data.

Before collecting data

Questions you need to ask yourself before collecting data?

- What data do I need to solve the problem?
- Where can I get this data from?
- Is the data available and accessible?
- Is the data labeled?
- Data should be up-to-date?

Type of data

- **Internal Logs / Systems**: Web traffic logs, Transaction histories, Sensor/IoT data.
- **External / Public Datasets**: Open repositories, Government and research datasets.
- **Synthetic Data**: Simulated environments[1].

[1] Useful when real data is scarce or privacy-restricted

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A company runs a large-scale network and wants to detect network intrusions (cyberattacks) in real-time.

You might collect:

- **Flow-level data**: Source IP, destination IP, ports, protocol, bytes transferred, duration, flags.
- **Packet-level data**: Packet timestamps, sizes, TCP flags.
- **Labels**: Whether each connection was **benign** or **malicious**.

Possible Data Sources:

- Network monitoring tools (e.g., Wireshark)
- Intrusion detection systems (IDS)
- Public datasets

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Data preparation

Data preprocessing, or data preparation, transforms the raw data from the previous step into clean data that is ready for analysis. Data preprocessing steps include:

- Data cleaning
- Data transformation
- Feature selection/exctraction
- Data labeling
- Data splitting
- Dealing with class imbalance

Poor data preparation can case bias in the predictions, poor generalization and training instability.

Data cleaning

Removes or fixes bad or inconsistent data.

- Missing Values:
  - Remove rows or columns with too many missing values;
  - Fill missing values using mean/median/mode, interpolation, or forward fill.
- Fixing Errors: Typos, out-of-range values, ...;
- Removing Duplicates;
- Filtering Outliers: Use statistical techniques (e.g., IQR) to identify and remove or flag extreme values.

Data transformation

Changes the format or structure of data to suit the model.

- Normalization / Scaling: Brings features to the same range (Important for some models)
- Encoding Categorical Variables : Label Encoding or One-Hot Encoding
- Binning / Discretization: Convert continuous variables into categories (e.g., packet size: small, medium, large)

Feature selection/exctraction

Choosing the most relevant features (or creating new ones) to improve model accuracy and reduce complexity.

- Remove irrelevant or redundant features;
- Select top features based on correlation importance, or statistical methods;
- Create new features (e.g., connection rate, average packet size)

Data labeling

For supervised learning, you need labeled data.
This may involve manual annotation, automated tools (like IDS logs), or matching logs with known ground truth.

Data splitting

Divide the data for training and evaluation.

- Training set: used to train the model;
- Validation set (optional): used to tune hyperparameters;
- Test set: used to evaluate final model performance.

Typical splits:
70% train, 15% validation, 15% test
80% train, 20% test

Dealing with class imbalance

Class can be imbalanced (common in intrusion detection, medical diagnosis, ...).
We can detect class imbalance with a bar chart of the target variable.

- Resampling techniques:
  - Randomly remove samples from the majority class (Simple, fast but risk of losing important information);
  - Duplicate or synthetically generate more minority samples - SMOTE (Synthetic Minority Over-sampling Technique)
- Class weighting: give more importance to the minority class during training.
- Anomaly detection methods : Train the model on only the majority class and use unsupervised models to flag anything "unusual".

Example

A company runs a large-scale network and wants to detect network intrusions (cyberattacks) in real-time.

Duration	**Protocol**	**Source IP**	**Destination IP**	**Bytes Sent**	**Label**
1.2 sec	TCP	192.168...	10.1...	4500	benign
0.3 sec	UDP	192.168...	10.2...	1100	attack

In this case data preparation can include:

- Droping or anonymization of IPs (not useful and may leak info);
- Encode protocol: converting categorical variables into a binary format;
- Normalization of **Duration** and **Bytes Sent**;
- Balance of the dataset if "attack" label is rare

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Data exploration and visualization[2]

It is the step where you get to know your data. It's about uncovering patterns, spotting problems, and forming hypotheses before you train any machine learning model. The goal is to:

- Understand the shape and structure of the dataset;
- Explore distributions of features (numerical and categorical)
- Discover correlations or patterns
- Compare class distributions (especially important in classification tasks)
- Guide feature selection and engineering
- Validate assumptions (e.g., normality, linearity)

[2] This step can be run in parallel with the Data preparation one

Data exploration and visualization

It includes:

- Summary Statistics: mean, std, class distribution, ...
- Visualizing Individual Features: Histograms, Boxplots, Bar Chart, Pie chart, ...
- Visualizing Relationships Between Features: Scatter plot, correlation matrix, ...

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- Summary Statistics:
  - `.describe()`
  - `.value_counts()`
  - `.info()`
- Visualizing Individual Features:

  Bar chart of Label column, ...
- Visualizing Relationships Between Features:

  Do some protocols (TCP/UDP/ICMP) correlate more with attacks?

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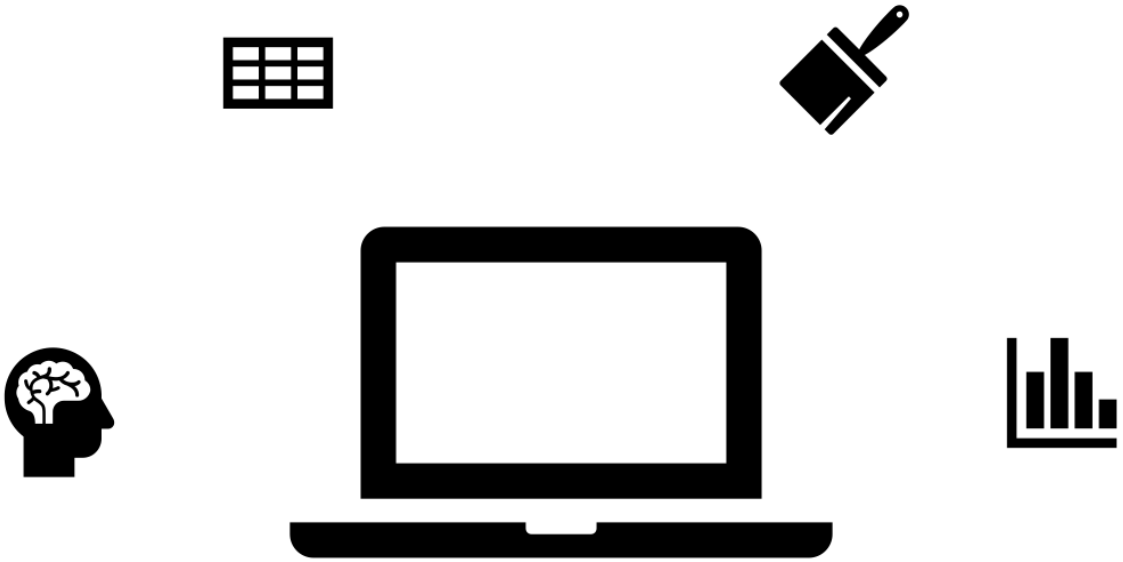
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